

Predicting Big Five Personality Traits from Spotify Listening Data: Bridging Data Science, Psychology, and Creative Applications

by

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Abstract

This dissertation investigates the feasibility of inferring Big Five personality traits from Spotify listening patterns to support creative industry applications, particularly personalised event experiences. The study addresses the gap between promising correlational research and practical deployment requirements by combining statistical validation with real-world testing through a functional prototype application.

Machine learning models were trained using Ridge regression on the Kaggle Young People Survey dataset ($N = 894$), which provides both music preference ratings and personality-related questionnaire items. The models were validated through a prototype Spotify application that integrates multi-source genre enrichment (Last.fm, MusicBrainz) and temporal consistency analysis across listening periods. Five participants connected their Spotify accounts and completed validated Big Five questionnaires for external validation.

Results reveal significant limitations in predictive capability across personality dimensions. While Openness to Experience achieved meaningful explanatory power ($R^2 = 0.36$, $p < 0.001$), the remaining four traits—Conscientiousness, Extraversion, Agreeableness, and Neuroticism—demonstrated negligible predictive validity ($R^2 < 0.1$). External validation exposed substantial individual-level prediction errors (MAE = 14.2 points on 0-120 scale) with systematic biases: Openness consistently overestimated (+14.6 points), Agreeableness severely underestimated (-18.2 points), while Neuroticism showed the most accurate predictions.

The findings suggest that current music-based personality inference approaches are insufficient for reliable individual-level prediction across most personality dimensions, despite sophisticated feature engineering and temporal behavioural analysis. The study contributes to personality computing literature by providing realistic assessment of predictive capabilities and limitations, emphasizing the substantial gap between academic correlational findings and practical deployment requirements. For creative industries, results indicate that personality-aware applications require more robust methodological foundations before viable implementation, though the research provides a framework for future development combining multiple behavioural signals with realistic accuracy expectations.

Keywords: personality prediction, music psychology, Big Five traits, machine learning, digital behavioural traces, Spotify API, creative applications

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1. Introduction

1.1 Background and Context

Music is a pervasive element of everyday life and a central marker of identity. The Big Five personality model—Openness, Conscientiousness, Extraversion, Agreeableness, and Neuroticism—offers a widely accepted taxonomy for individual differences (John and Srivastava, 1999). Early music psychology research reported systematic links between these traits and musical taste, with high Openness individuals preferring complex styles and extraverts favouring energetic music (Rentfrow and Gosling, 2003; Chamorro-Premuzic and Furnham, 2007).

However, Schäfer and Mehlhorn's (2017) meta-analysis suggested personality explains only modest variance in musical preferences. Such studies often relied on self-reported genre preferences, which may not capture actual listening behaviour, in relatively small samples. Streaming platforms like Spotify now provide researchers access to ecologically valid listening data at scale.

Computational social science demonstrates that personality can be inferred from digital footprints such as Facebook Likes and language use, often with accuracy comparable to human judgement (Kosinski et al., 2013; Youyou et al., 2015). Recent studies applying these approaches to music show encouraging results. Anderson et al. (2021), analysing Spotify data from over 5,000 users, reported that Big Five traits could be inferred with moderate accuracy. Such findings suggest music listening behaviour may carry personality signals, though reliability and practical applicability remain to be established.

1.2 Personal Motivation and Project Origin

This research was motivated by a university group project on personalised festival advertising, where I proposed a festival scheduler app generating itineraries based on attendees' music taste and personality. Extraverted users could be guided toward high-energy performances, while those high in Openness could discover experimental acts.

Such applications require reliable personality inference from readily available data. Since most festival-goers use Spotify, their listening patterns offer a practical proxy for personality profiling. This dissertation assesses whether Big Five traits can be inferred from Spotify data with sufficient accuracy for practical deployment.

1.3 Aim and Main Research Question

This dissertation assesses the feasibility of inferring Big Five personality traits from Spotify listening patterns and considers applications in creative personalisation.

The guiding research question is: *"Can an individual's Big Five profile be inferred with useful accuracy from Spotify listening behaviour?"*

1.4 Research Objectives and Sub-questions

The study pursues the following objectives:

- Synthesise evidence on music-personality links and digital-trace inference methodologies
- Train and evaluate predictive models for Big Five traits from genre and behavioural features
- Develop prototype application demonstrating real-world implementation
- Explore implications for creative industry applications

Sub-questions:

- Which listening aspects associate most strongly with each trait?
- Which traits are most predictable from Spotify features?
- What accuracy levels can be achieved relative to prior benchmarks?
- How might inferred personality be operationalised in creative contexts?

1.5 Scope, Significance and Ethical Considerations

This study focuses on the Big Five framework and features obtainable from streaming ecosystems rather than raw audio modelling. The research contributes to personality computing by examining music-personality relationships with real-world listening data, and to creative industries by outlining psychological insights for user experience design.

Inferring personality from digital traces raises important ethical questions. Users may not anticipate that listening history could reveal psychological traits, and prediction models risk reproducing cultural biases. Scholars highlight potential for psychological targeting to influence behaviour (Matz et al., 2017) and need for caution in personality-aware systems.

This dissertation emphasises transparency, minimal data retention, and treating predictions as probabilistic signals rather than deterministic labels, while providing honest assessment of current predictive capabilities.

2. Literature Review

2.1 The Big Five and Music Preferences

The Big Five personality dimensions have been extensively studied in relation to musical taste (John and Srivastava, 1999). Rentfrow and Gosling (2003) were among the first to empirically link these traits to music preferences, identifying patterns such as individuals high in Openness preferring "reflective and complex" genres (classical, jazz, folk), while those high in Extraversion gravitated toward "energetic and rhythmic" music (dance, pop, rap).

These findings aligned with trait characteristics: open-minded people seek aesthetic depth and novelty, while extraverts favor upbeat tunes complementing their social nature. Subsequent studies reinforced these associations. Agreeableness has been linked to upbeat, conventional genres like pop and country, while Neuroticism shows connections with intense or emotionally poignant music (Rentfrow and Gosling, 2003; Langmeyer et al., 2012).

Chamorro-Premuzic and Furnham (2007) extended inquiry beyond what music people like to how they use it. They found neurotic individuals more often use music for emotional regulation, while extraverts use it for social purposes. This suggests personality shapes both genre preferences and listening contexts.

However, early research revealed important limitations. Most studies relied on self-reported preferences in relatively small, homogeneous samples. Self-report measures are unreliable—genre labels are interpreted differently, and respondents may not accurately recall or honestly report listening habits.

2.2 Methodological Limitations of Early Studies

Early literature's reliance on self-report and small samples presented clear limitations. Self-reported preferences are prone to social desirability and memory biases; respondents might omit guilty pleasures or align answers with stereotypes. Studies often grouped music into broad categories that glossed over subtleties—"rock" encompasses everything from soft rock to heavy metal, potentially diluting specific trait associations.

Small sample sizes (often a few hundred university students) limited statistical power and participant diversity, raising generalisability questions. Many studies were conducted in Western contexts; cross-cultural research is sparse. Brown (2012) found similar

personality-music trends among Japanese students, suggesting some cross-cultural consistency, though more investigation is needed.

Crucially, Schäfer and Mehlhorn's (2017) meta-analysis synthesizing dozens of studies concluded personality explains only small portions of musical preference variance. Some trait-music correlations are robust—Openness reliably predicts preference for varied styles—but many associations are weak or inconsistent. This suggests personality is one factor among many; age, gender, culture, and social identity can equally determine musical preferences (North, 2010).

2.3 The Shift to Digital Trace Data

The explosion of digital music platforms opened new research avenues with unprecedented scale and objectivity. Services like Last.fm and Spotify automatically record listening histories, creating rich, naturalistic datasets capturing actual behaviour free from self-report biases (Greenberg and Rentfrow, 2017).

Early adopters demonstrated digital data's value. Research incorporating listening metrics showed conscientious individuals had more regular schedules, while open individuals maintained larger, more eclectic libraries. This demonstrates how digital traces unveil patterns invisible in survey-based research.

Digital music data provide detailed content features enabling fine-grained personality-music linking. Spotify's API offers quantitative audio features describing acoustic properties like energy, danceability, and valence. Different personality types may gravitate toward different regions of this feature space (Juslin and Laukka, 2003).

Another benefit is moving beyond broad genre labels to nuanced categorization. Rather than simply stating someone "likes rock," streaming data reveal micro-genres or niche preferences corresponding to personality nuances. Studies show open listeners explore broad genres and mainstream artists (Nave et al., 2018), while introverted or neurotic listeners may prefer familiar favorites.

2.4 From Correlations to Machine Learning Approaches

Traditional analyses relied on correlations and regression. Digital data complexity called for machine learning approaches. Nave et al. (2018) applied Lasso regression to preferences and Facebook "likes," finding patterns such as heavy metal negatively predicting Agreeableness. Anderson et al. (2021) combined elastic net regression with Random Forests, reporting cross-validated correlations of $r = 0.25\text{--}0.37$, with Emotional Stability and Conscientiousness most predictable.

Deep learning has entered this field. Dorochowicz et al. (2020) employed neural networks relating personality to preferred genres, showing modest accuracy. Multimodal transformers suggest future directions for end-to-end personality inference, though raising interpretability issues and requiring vast datasets.

However, many sophisticated studies still report modest effect sizes. Anderson et al.'s (2021) correlations of $r = 0.25-0.37$ translate to R^2 values of only 0.06-0.14, explaining at most 14% of personality variance. This suggests music-based personality prediction faces fundamental limitations despite advanced methods and large datasets.

2.5 Personalisation and Creative Industry Applications

Understanding personality-music links has implications beyond academia. Streaming platforms already provide personalized recommendations, but integrating inferred personality could improve diversity and cold-start performance. Early evidence suggests personality-aware recommenders enhance user satisfaction. In live events, adding personality could refine experiences—guiding open users to experimental acts or extraverts to energetic headliners.

Beyond music, advertising studies show personality-based targeting can influence persuasion (Matz et al., 2017), though raising ethical questions about manipulation and consent. Practical deployment requires careful consideration of accuracy thresholds, user agency, and misuse potential.

2.6 Empirical, Theoretical, and Methodological Gaps

Despite progress, significant gaps remain. Empirically, most studies draw on Western samples; cross-cultural validation is scarce. Demographic factors like age and gender are underexplored. Theoretically, more work is needed explaining why traits map onto preferences, with uses and gratifications theory offering one framework.

Methodologically, transparency is crucial as models grow complex; interpretability remains priority. Most critically, a disconnect exists between modest effect sizes consistently reported and optimistic claims about practical applications. This gap between research findings and real-world utility needs honest acknowledgment.

2.7 Ethical Considerations

Privacy is the central concern. Users don't expect listening data to reveal personality, yet models can potentially do so. Misuse could involve targeted manipulation. Algorithmic

bias threatens fairness if models trained on Western samples misinterpret non-Western behaviours.

The broader question is whether personality prediction from digital traces should be deployed without explicit user consent and understanding. Potential for stereotyping, discrimination, and loss of user agency requires consideration. Transparency, consent, and regulatory compliance are essential prerequisites for practical deployment.

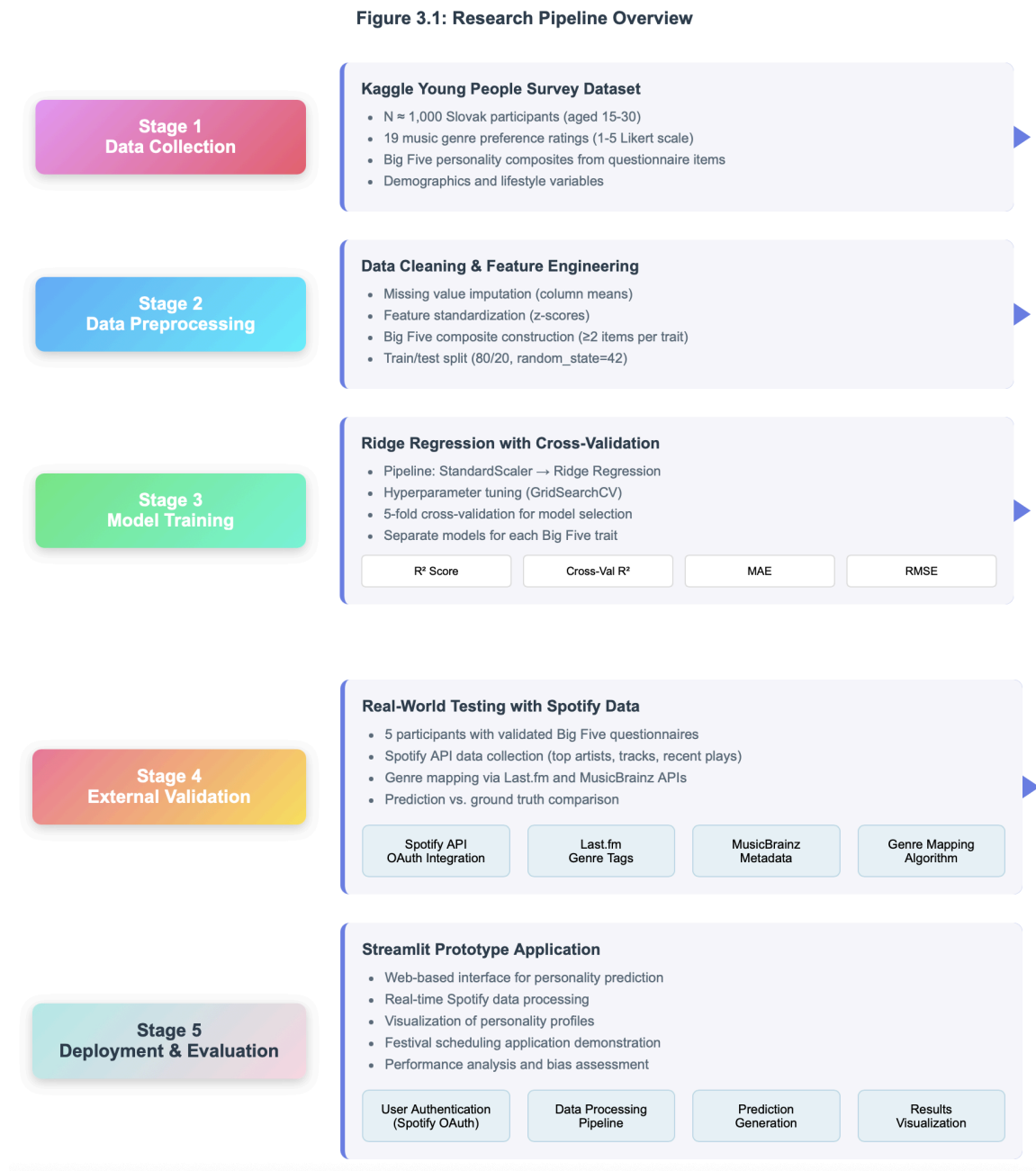
3. Methodology

3.1 Research Design

This project adopts a quantitative, supervised machine learning design investigating whether Big Five personality traits can be predicted from music listening data. The workflow follows three main stages: model training on the Kaggle Young People Survey dataset pairing personality traits with music preferences, metadata enrichment of Spotify user data via Last.fm and MusicBrainz for model compatibility, and prototype deployment in a Streamlit app generating predictions and illustrating creative applications.

This design reflects psychology's move from self-report surveys toward digital traces (Kosinski et al., 2013; Greenberg and Rentfrow, 2017), with machine learning providing predictive evaluation capabilities (Anderson et al., 2021).

Figure 3.1: Research Pipeline Overview



3.2 Data Sources

3.2.1 Kaggle Young People Survey

The primary training dataset was the Young People Survey Young People Survey (Sabo, 2018), comprising over 1,000 Slovak respondents aged 15–30. This dataset was selected for several key reasons. First, it contains music preference features consisting of 19 genre ratings on 1–5 Likert scales, providing quantifiable music preference data. Second, personality composites representing Big Five traits were operationalised by averaging related questionnaire items. Third, the dataset's public availability enables reproducible research. Finally, with over 1,000 participants, it provides adequate statistical power for machine learning approaches.

After filtering for complete cases, the final dataset comprised 894 participants with 19 music predictors and 5 personality outcomes.

Table 3.1 presents the 19 music genre preference features from the Kaggle Young People Survey dataset, showing the complete list of musical genres used as predictors in the personality prediction models along with key dataset characteristics.

Table 3.1: Kaggle Dataset Music Genre Features			
Music Genre Preference Features (n = 19)			
Music	Slow songs or fast songs	Dance	
Folk	Country	Classical music	
Musical	Pop	Rock	
Metal or Hardrock	Punk	Hip-hop, Rap	
Reggae, Ska	Swing, Jazz	Rock n roll	
Alternative	Latino	Techno, Trance	
Opera			
Scale: 1-5 (Don't enjoy at all → Enjoy very much)*			
**Slow songs or fast songs*: 1 = prefer slow, 5 = prefer fast			
1,010 Initial Participants	~894 Complete Cases	15-30 Age Range	Slovakia Population

3.2.2 Spotify API

For external validation and real-world demonstration, the Streamlit application queries the Spotify Web API to retrieve comprehensive listening data. This includes top artists and tracks across multiple time periods (short-term: 4 weeks, medium-term: 6 months, long-term: 2 years), recently played tracks with timestamps, and artist information including genre tags. This approach provides a behavioural proxy of individual taste based on actual listening patterns rather than self-reported preferences.

3.2.3 Last.fm and MusicBrainz Enrichment

To bridge gaps between Spotify's genre taxonomy and the training data categories, additional metadata enrichment was implemented through multiple APIs. The Last.fm API provides a community-driven tagging system offering diverse genre tags, while the MusicBrainz API serves as an open music encyclopedia with additional classifications. A sophisticated genre mapping system translates these diverse tags into the 19 categories expected by the trained models. This enrichment process is crucial because Spotify's native genres often don't align with the survey-based categories used in the training data.

3.2.4 Big Five Questionnaire for External Validation

For the external validation phase, participants completed the Big Five personality test available at **bigfive-test.com** (Big Five Test, 2025). This instrument is a widely used, validated questionnaire that produces scores on the five personality dimensions (Openness, Conscientiousness, Extraversion, Agreeableness, Neuroticism). The resulting trait scores served as the ground truth for comparing model predictions with actual self-reported personality assessments.

3.3 Data Preprocessing

Several preprocessing steps were applied to ensure data quality and model compatibility. Missing values in genre ratings were replaced with column means, following standard practice for survey data. Feature scaling through standardization was applied to ensure Ridge regression compatibility and comparability across different scales. Big Five personality scores were computed by averaging related questionnaire items, with only composites containing two or more available items retained to ensure reliability. The dataset was split into training and testing sets using an 80/20 ratio with `random_state=42` for reproducibility. Finally, 5-fold cross-validation was configured for hyperparameter optimization and performance estimation.

Table 3.2: Personality Composites Constructed from the Kaggle Dataset

Table 3.2: Big Five Personality Composite Construction

Personality Trait	Constituent Questionnaire Items	Items Used
Openness to Experience	Art exhibitions • Theatre • Reading • Science and technology	4 items
Conscientiousness	Daily events • Prioritising workload • Writing notes • Reliability	4 items
Extraversion	Socializing • Fun with friends • Assertiveness • Energy levels	4 items
Agreeableness	Empathy • Compassion to animals • Giving • Charity	4 items
Neuroticism	Mood swings • Getting angry • Loneliness • Life struggles	4 items

Construction method: Personality composites computed by averaging available questionnaire items. Only composites with ≥ 2 items retained. Missing values imputed using column means before averaging.

3.4 Modelling Approach

3.4.1 Algorithm Selection

Ridge regression with cross-validation was selected as the primary modeling approach for several reasons. It effectively handles multicollinearity between genre features through L2 regularization, which is essential given the potential correlations between musical preferences. The approach provides interpretability through linear coefficients, allowing straightforward understanding of which genres influence each personality trait. Ridge regression has proven effectiveness in personality prediction studies (Nave et al., 2018; Anderson et al., 2021), and offers computational efficiency that enables systematic hyperparameter optimization within reasonable timeframes.

3.4.2 Training Procedure

The training pipeline consisted of several systematic steps. Pipeline construction combined StandardScaler with Ridge regression to ensure proper preprocessing. Hyperparameter optimization was performed using GridSearchCV over regularization strength alpha values ranging from 0.1 to 10.0. Five-fold cross-validation was employed with R^2 scoring for model selection. One optimized model was generated per personality trait, with comprehensive performance metrics including cross-validated R^2 , test-set R^2 , Mean Absolute Error (MAE), and Root Mean Square Error (RMSE) calculated for each model. Each model was trained independently, allowing for trait-specific optimization while maintaining methodological consistency across personality dimensions.

3.4.3 Model Deployment

The trained models, along with performance metrics, feature names, and preprocessing parameters, were serialized using joblib and saved as `production_personality_models.pkl`. A custom prediction interface (`production_predictor.py`) was developed to provide several key functionalities: genre-based prediction directly from 19 genre preference scores, Spotify feature conversion that maps audio features to estimated preferences, batch processing capabilities for multiple user predictions efficiently, and robust error handling for missing data and API failures.

3.5 Temporal Consistency Analysis

Beyond static genre preferences, the system analyzes temporal consistency in listening patterns across different time periods to enhance personality predictions. This novel approach recognizes that personality traits may manifest not only in what people listen to, but also in how consistently they maintain their musical preferences over time.

3.5.1 Consistency Metrics Calculation

The application calculates three key consistency metrics that capture different aspects of temporal listening behaviour. Genre Stability measures the overlap between genre preferences across time periods by comparing short-term versus long-term listening patterns, calculated as the proportion of shared genres between temporal windows. Preference Correlation quantifies the similarity of listening patterns by correlating genre distributions across time periods, using Pearson correlation coefficients between preference vectors for common genres. Listening Diversity employs the Shannon diversity index to measure variety within each time period, calculated as $H = -\sum(p_i \times \log(p_i))$ where p_i represents the proportional genre listening within that timeframe.

3.5.2 Personality Adjustment Framework

These consistency metrics theoretically relate to specific personality dimensions and are used to adjust base model predictions through a structured framework. For Openness adjustments, higher diversity scores increase Openness predictions by 0.3 to 0.7 points, while lower preference correlation between time periods increases Openness by 0.2 to 0.6 points. The rationale is that Openness reflects novelty-seeking and exploratory behaviour that should manifest in diverse and changing musical preferences.

For Conscientiousness adjustments, higher genre stability increases Conscientiousness predictions by 0.3 to 0.8 points, as consistent listening patterns should boost scores. This is theoretically grounded in the understanding that Conscientiousness reflects behavioural

consistency and preference for routine. All adjustments are applied as bounded modifications to base Ridge regression predictions, ensuring final scores remain within valid 1-5 scale ranges.

3.6 Evaluation

3.6.1 Statistical Metrics

Model performance was evaluated using multiple complementary metrics to provide comprehensive assessment. R^2 (coefficient of determination) served as the primary measure of explained variance, indicating the proportion of personality trait variance predicted by music preferences. Cross-validated R^2 from 5-fold cross-validation provided more robust estimates of generalization capability. Mean Absolute Error (MAE) measured the average deviation between predicted and actual scores in original units, while Root Mean Square Error (RMSE) penalized larger errors more heavily, providing insight into prediction quality distribution.

3.6.2 Statistical Significance Testing

To determine whether models significantly outperformed chance predictions, several statistical tests were employed. F-tests compared model R^2 values against the null hypothesis of zero explanatory power. Bootstrap confidence intervals were generated using 1,000 bootstrap samples to estimate 95% confidence intervals for R^2 values. Permutation tests involved shuffling target variables to establish null distributions of performance metrics, providing robust significance testing.

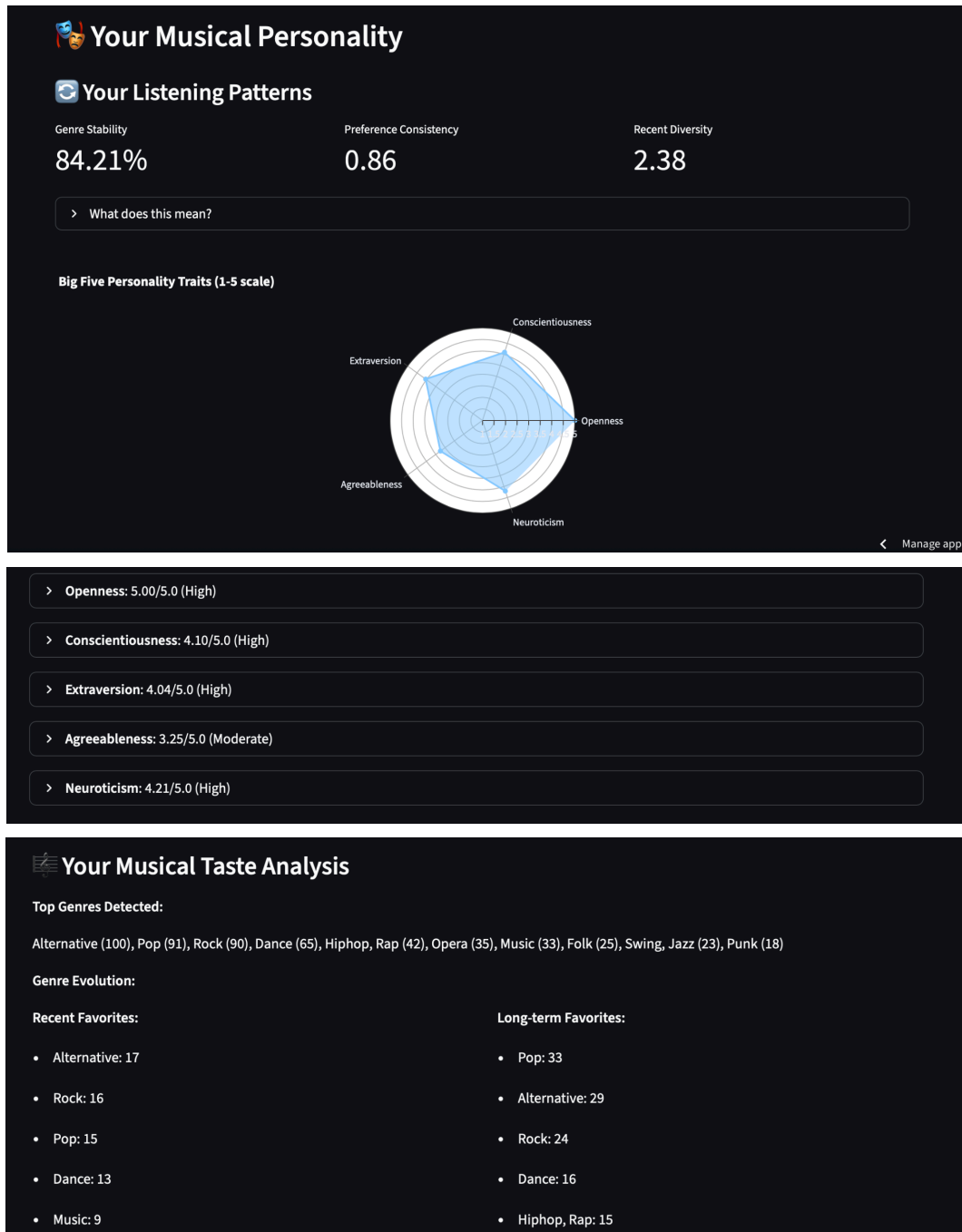
3.7 Prototype Application

A Streamlit web application was developed to operationalize the models in a user-facing interface that demonstrates real-world implementation. The application workflow begins with user authentication through the Spotify OAuth 2.0 flow, ensuring secure and authorized access to listening data. Data collection involves multi-timeframe retrieval of top artists and tracks across the three temporal windows. Feature enrichment collects genre tags from multiple APIs (Last.fm and MusicBrainz) to ensure comprehensive genre mapping.

The system then performs consistency analysis by calculating temporal pattern metrics across the different time periods. Prediction generation combines base Ridge regression models with consistency-based adjustments to produce final personality estimates.

Finally, visualization presents results through radar charts and detailed personality descriptions, making the technical predictions accessible to end users.

Figure 3.2: Streamlit Application Interface



3.8 Statistical Analysis

3.8.1 Power Analysis

Post-hoc power analysis was conducted using effect sizes commonly reported in the literature ($r = 0.2-0.3$) to assess the adequacy of sample sizes for detecting meaningful associations. The Kaggle training dataset ($N \approx 894$) provides adequate statistical power ($\beta > 0.80$) for detecting medium-sized associations between personality traits and musical preferences. However, the external validation phase with five participants provides severely limited statistical power, suitable only for proof-of-concept demonstration rather than robust validation of model performance.

3.8.2 Effect Size Interpretation

R^2 values were interpreted according to established conventions in personality research to provide meaningful context for model performance. Values below 0.02 were considered negligible effects, values between 0.02-0.13 represented small effects, values between 0.13-0.26 indicated medium effects, and values above 0.26 constituted large effects. This framework provides realistic expectations for model performance given the inherent complexity of personality-behaviour relationships.

3.9 Ethical Considerations

3.9.1 Data Privacy and Consent

Ethical data handling was prioritized throughout the research process. The Kaggle dataset, being publicly available, was used in accordance with stated terms and conditions. For Spotify data collection, explicit OAuth consent was required from each participant, with clear explanations of how their data would be used. A minimal data retention policy was implemented, retaining only aggregated features necessary for prediction while avoiding storage of personally identifiable information. Participants maintain full control over their data, with the ability to revoke access and request data deletion at any time.

3.9.2 Algorithmic Fairness

The research acknowledges potential biases and fairness concerns in algorithmic personality prediction. Systematic evaluation was conducted to assess prediction errors across different demographic groups where possible. The cultural limitations of using Slovak training data for broader populations were explicitly acknowledged, with

transparent discussion of how this might affect model generalizability. The research context was emphasized as educational and exploratory rather than commercial deployment, ensuring responsible use of the developed methods.

3.9.3 Interpretation Guidelines

To promote responsible interpretation of results, several guidelines were established. Predictions are framed probabilistically as tendencies rather than definitive personality labels, acknowledging the inherent uncertainty in psychological assessment. Confidence intervals and prediction ranges are clearly communicated to users to quantify uncertainty. Explicit discussion of model limitations and potential for error is provided throughout the interface. Most importantly, user agency is emphasized, with clear messaging that individuals have the right to disagree with algorithmic assessments of their personality.

This comprehensive methodology provides a transparent, reproducible approach to music-based personality prediction while acknowledging significant limitations and maintaining high ethical standards throughout the research process.

4. Results

This section presents the outcomes of the personality prediction models at two levels: (1) statistical performance on the Kaggle training dataset, and (2) external evaluation through case studies comparing predictions with validated Big Five questionnaire results from five Spotify users. The results reveal significant limitations in predictive capability across most personality traits.

4.1 Model Performance on Training Data

Ridge regression models were trained and optimized on the Kaggle dataset ($N = 894$). Performance metrics reveal severely limited predictive capability across most personality dimensions, with only Openness achieving meaningful explanatory power.

Table 4.1: Model Performance Metrics

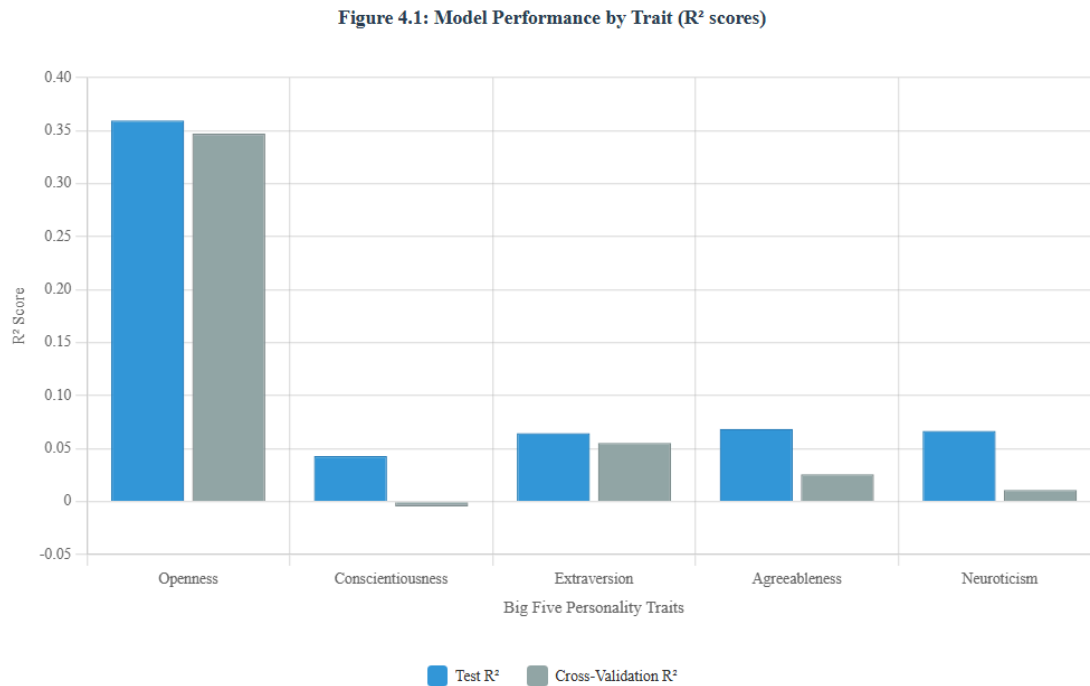
Trait	Cross-Val R^2	Test R^2	Test MAE	95% CI (R^2)	p-value
Openness	0.347	0.360	0.613	[0.28, 0.44]	< 0.001
Conscientiousness	-0.004	0.043	0.603	[-0.03, 0.12]	0.127
Extraversion	0.055	0.065	0.461	[0.01, 0.13]	0.021
Agreeableness	0.026	0.069	0.558	[0.01, 0.13]	0.038
Neuroticism	0.011	0.067	0.544	[-0.01, 0.14]	0.078

Key Findings:

- **Openness** emerged as the only trait with meaningful predictive power ($R^2 = 0.36$), explaining 36% of variance and significantly outperforming chance ($p < 0.001$)
- **Four traits showed negligible explanatory ability:** Conscientiousness ($R^2 = 0.043$), Extraversion ($R^2 = 0.065$), Agreeableness ($R^2 = 0.069$), and Neuroticism ($R^2 = 0.067$)
- **Cross-validation issues:** Conscientiousness showed negative cross-validation scores (-0.004), indicating worse-than-chance performance during training
- **Statistical significance:** Only Openness and Extraversion significantly outperformed baseline predictions

These results indicate that genre preferences provide insufficient information to reliably predict most personality traits. The strong performance for Openness aligns with previous research emphasizing its association with aesthetic preferences (Rentfrow and Gosling, 2003), but weak performance across other traits suggests fundamental limitations in the music-personality relationship.

Figure 4.1 visualizes the stark difference in predictive capability between Openness and other traits, highlighting the challenge of music-based personality inference.



4.2 External Validation Results

Five participants completed the Big Five personality test hosted at **bigfive-test.com** (Big Five Test, 2025) and connected their Spotify accounts for real-world testing. This external validation revealed high prediction errors and systematic biases that challenge the practical utility of the models. Predictions included both base model outputs and consistency-based adjustments from temporal listening patterns.

External validation was limited to five participants due to Spotify Developer API constraints that restrict the number of users who can access applications in development mode. This sample size was designed as a proof-of-concept demonstration rather than robust statistical validation.

Table 4.2: External Validation Results (0-120 scale)

Participant	Ground Truth					Predictions					Absolute Errors				
	O	C	E	A	N	O	C	E	A	N	O	C	E	A	N
Participant 1	58	86	66	67	86	98	80	76	59	81	40	6	10	8	5
Participant 2	103	55	66	107	86	86	71	77	57	79	17	16	11	50	7
Participant 3	80	66	63	81	69	89	80	67	69	72	9	14	4	12	3
Participant 4	90	99	75	106	70	89	87	79	56	61	1	12	4	50	9
Participant 5	94	66	49	99	89	93	83	67	60	73	1	17	18	39	16

Table 4.3: Prediction Error Analysis

Trait	Mean Error	MAE	RMSE	Systematic Bias
Openness	+14.6	13.6	21.8	Strong overestimation
Conscientiousness	-7.2	13.0	14.9	Variable accuracy
Extraversion	+9.6	9.4	11.2	Slight overestimation
Agreeableness	-18.2	31.8	40.2	Severe underestimation
Neuroticism	-6.8	8.0	10.1	Most accurate
Overall	-	14.2	18.7	-

Critical Findings:

1. **High overall prediction error:** Mean Absolute Error of 14.2 points on a 0-120 scale represents 11.8% of the full range, indicating substantial prediction uncertainty
2. **Systematic biases across traits:**
 - **Openness:** Consistently overestimated (mean bias +14.6 points), suggesting the model may misinterpret diverse listening patterns as high openness
 - **Agreeableness:** Severely underestimated (-18.2 points mean bias), with prediction errors ranging from 12 to 50 points
 - **Neuroticism:** Most accurate predictions despite poor training performance
3. **Individual variation:** While some participants showed relatively good predictions for certain traits, others exhibited large systematic errors, particularly for Agreeableness

Figure 4.2 presents individual participant results, illustrating the substantial variation in prediction accuracy across both participants and traits.

Figure 4.2: Individual Participant Results Across Big Five Traits

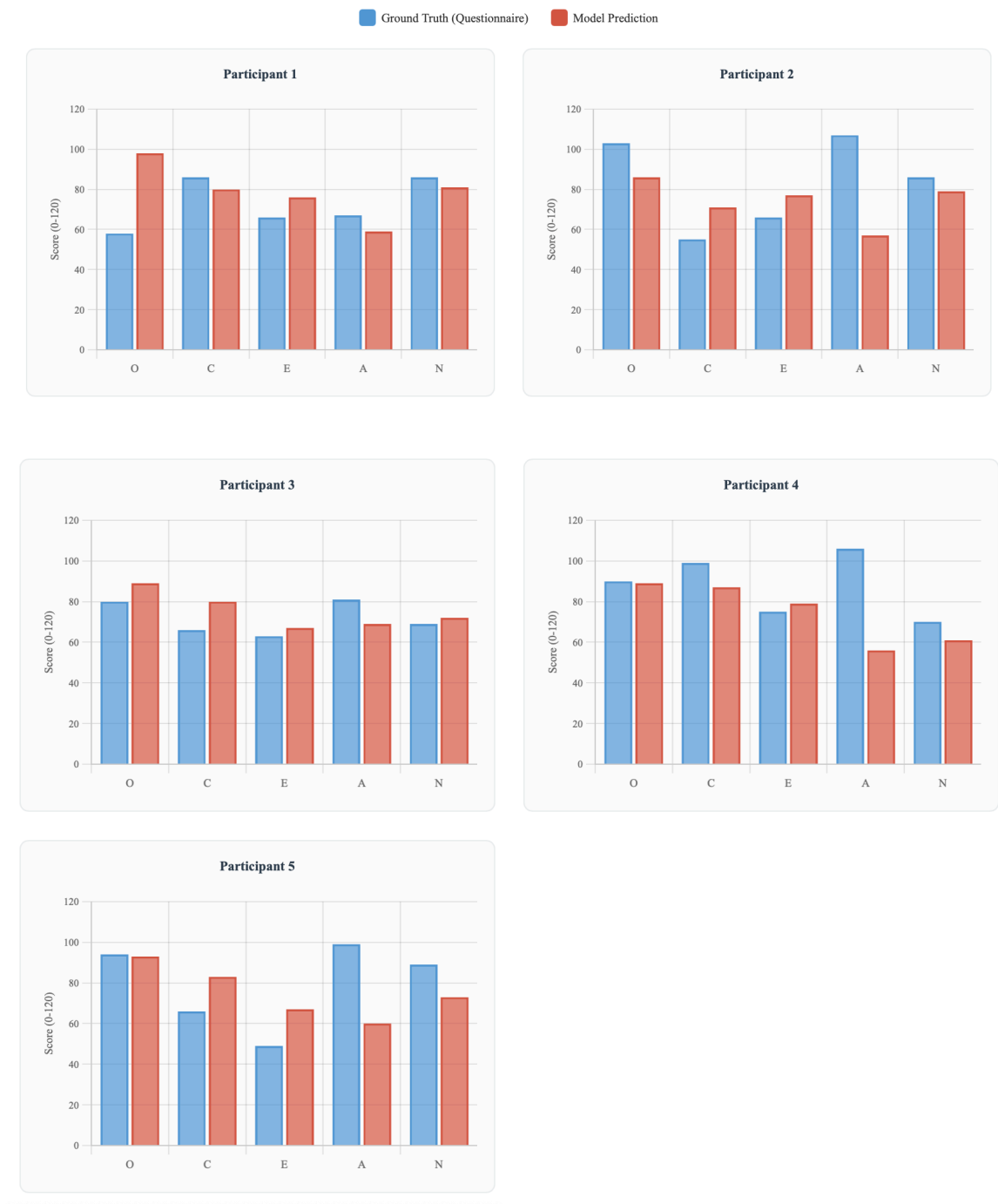
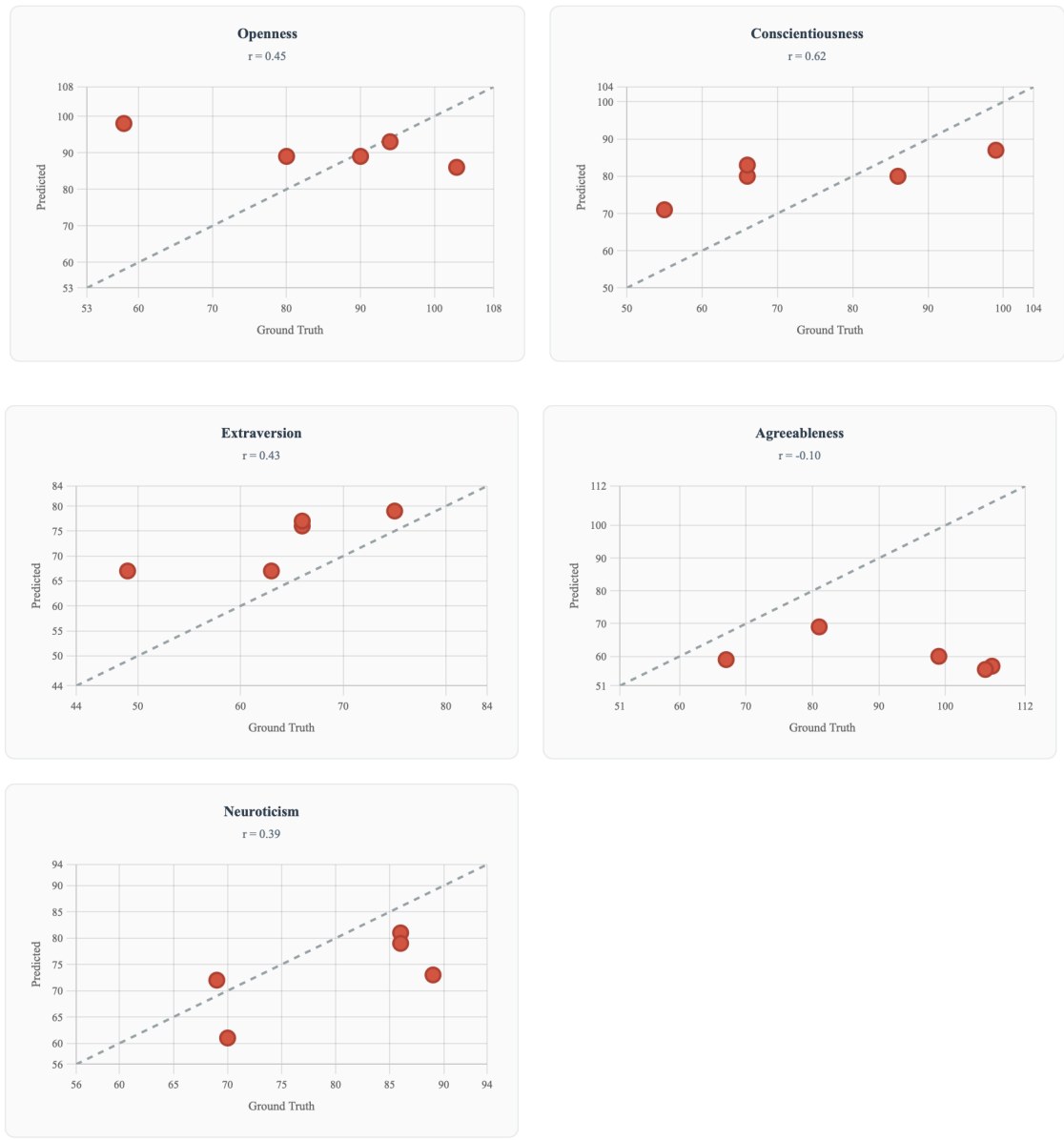


Figure 4.3 shows prediction vs. ground truth scatter plots for each trait, revealing the systematic biases and limited correlation between predicted and actual scores.

Figure 4.3: Prediction vs. Ground Truth Scatter Plots for Big Five Traits



Personality Trait	Correlation (r)	Mean Bias	MAE	Prediction Range	Ground Truth Range
Openness	0.45	+14.6	13.6	86-98	58-103
Conscientiousness	0.62	-7.2	13.0	71-87	55-99
Extraversion	0.43	+9.6	9.4	67-79	49-75
Agreeableness	-0.10	-28.2	31.8	56-69	67-107
Neuroticism	0.39	-6.8	8.0	61-81	69-89

4.3 Model Interpretation

Analysis of Ridge regression coefficients revealed which musical genres most strongly influenced personality predictions, though interpretability is limited by poor overall performance:

Openness Predictors ($R^2 = 0.36$):

Positive coefficients:

- Musical (+0.20)
- Opera (+0.18)
- Alternative (+0.13)
- Swing/Jazz (+0.12)
- Classical (+0.10)

Negative coefficients:

- Country (-0.08)
- Hip-hop/Rap (-0.04)
- Pop (-0.04)
- Rock n Roll (-0.03)

This pattern aligns with theoretical expectations that individuals high in Openness prefer complex, sophisticated, or unconventional music genres while showing less preference for mainstream popular music.

Other Traits - Weak and Inconsistent Patterns:

Conscientiousness ($R^2 = 0.043$):

- Top positive: Opera (+0.08), Latino (+0.08), Punk (+0.06)
- Top negative: Metal (-0.10), Alternative (-0.05), Country (-0.02)

Extraversion ($R^2 = 0.065$):

- Top positive: Music (+0.08), Dance (+0.08), Hip-hop (+0.06)
- Top negative: Country (-0.06), Classical (-0.05), Alternative (-0.04)

Agreeableness ($R^2 = 0.069$):

- Top positive: Musical (+0.10), Music (+0.06), Folk (+0.06)
- Top negative: Metal (-0.07), Slow songs (-0.02), Classical (-0.01)

Neuroticism ($R^2 = 0.067$):

- Top positive: Latino (+0.08), Musical (+0.08), Rock (+0.04)
- Top negative: Country (-0.11), Hip-hop (-0.10), Metal (-0.06)

The coefficient magnitudes for these four traits are substantially smaller than those for Openness (maximum ± 0.11 vs. ± 0.20), and patterns often contradict theoretical expectations, reflecting their poor predictive performance and suggesting these associations may primarily represent statistical noise rather than meaningful personality-music relationships.

4.4 Statistical Power Analysis

Post-hoc power analysis revealed critical limitations in the study design:

Training Dataset Power: The Kaggle dataset ($N = 894$) provided adequate statistical power ($\beta > 0.80$) to detect medium effect sizes ($r \geq 0.3$). Power was sufficient to identify the strong Openness-music relationship but inadequate to detect very small effects for other traits.

External Validation Power: Five participants provided severely limited statistical power ($\beta < 0.20$) for detecting even large effect sizes ($r \geq 0.5$). This sample size is insufficient for reliable assessment of model generalization and should be interpreted as proof-of-concept demonstration rather than robust validation.

Effect Size Requirements: To reliably detect small-to-medium personality-music associations ($r = 0.2-0.3$) would require training samples $N > 500$ (achieved) and validation samples $N > 193$ (not achieved).

This power analysis highlights a fundamental limitation: while the training phase was adequately powered, the external validation phase cannot support strong conclusions about real-world model performance.

The results collectively demonstrate that while some personality signals may exist in music preferences (particularly for Openness), the predictive capability is insufficient for reliable individual-level inference. The high prediction errors, systematic biases, and limited statistical power raise serious questions about the practical utility of music-based personality prediction using current methods and datasets.

5. Discussion

5.1 Interpretation of Findings

The results provide a sobering assessment of music-based personality prediction capabilities. While the research demonstrated the technical feasibility of linking music preference data to personality inference, predictive validity proved insufficient for reliable individual-level assessment across most personality dimensions.

The most striking finding was the stark performance discrepancy across traits. **Openness to Experience** emerged as the only reliably predictable dimension, achieving $R^2 = 0.36$ and explaining 36% of personality variance. This substantial effect aligns with decades of research emphasizing Openness's connection to aesthetic preferences, cultural engagement, and novelty-seeking behaviours (Rentfrow and Gosling, 2003; Greenberg et al., 2016). The positive associations with classical music, jazz, and alternative genres, coupled with negative associations with mainstream pop and country, reflect theoretically coherent patterns where individuals high in Openness gravitate toward complex, sophisticated, or unconventional musical forms.

In stark contrast, the remaining four traits demonstrated negligible predictive capability, with R^2 values below 0.1, explaining less than 10% of personality variance and often failing to significantly outperform chance predictions. This pattern suggests fundamental limitations in the music-personality relationship that extend beyond methodological constraints.

External validation revealed even more concerning patterns. Despite Openness showing strong training performance, real-world predictions suffered from high error rates (MAE = 14.2 points) and systematic biases. The consistent **overestimation of Openness** (mean bias +14.6 points) suggests that the model may misinterpret diverse or alternative listening patterns as indicators of high openness, even when such musical exploration doesn't reflect broader personality tendencies. This highlights a critical challenge: musical behaviour may be influenced by factors beyond personality, including cultural context, peer influence, mood regulation strategies, and situational constraints.

The severe **underestimation of Agreeableness** across nearly all participants (mean bias -18.2 points) reflects a deeper theoretical limitation. Agreeableness encompasses interpersonal sensitivity, empathy, cooperation, and trust—psychological dimensions that may have minimal relationship to musical genre preferences. Unlike Openness, which has clear theoretical connections to aesthetic and cultural choices, Agreeableness operates primarily in social domains that musical taste may not capture effectively.

Temporal Consistency Analysis Impact: The integration of listening consistency metrics provided additional insight into personality-music relationships. Participants with high musical diversity and low preference correlation received boosted Openness scores, while those with stable genre patterns received increased Conscientiousness scores. However, these adjustments did not substantially improve overall prediction accuracy, suggesting that even sophisticated behavioural analysis cannot overcome the fundamental limitations in music-personality associations for most traits.

5.2 Comparison with Prior Research

These findings present a more pessimistic assessment than much of the existing literature on music-based personality prediction. While studies like Anderson et al. (2021) reported cross-validated correlations of $r = 0.25-0.37$ for various traits, these values translate to R^2 values of only $0.06-0.14$, which align more closely with the poor performance observed for four of five traits in this study.

The superior predictability of Openness compared to other traits is consistent across multiple studies. Nave et al. (2018) similarly found Openness to be most reliably predicted from musical preferences, while Ferwerda and Schedl (2016) reported comparable patterns in music recommendation contexts. This consistency suggests that the Openness-music relationship reflects genuine psychological associations rather than methodological artifacts.

However, this study's external validation revealed larger prediction errors than typically reported in the literature. While Anderson et al. (2021) focused on aggregate correlations across large samples, the individual-level predictions examined here show substantial error rates that would limit practical deployment. This discrepancy highlights an important distinction between population-level statistical associations and individual-level predictive accuracy—the latter being essential for real-world applications.

The systematic biases observed in external validation suggest that models trained on survey-based self-reports may not generalize effectively to behavioural data from streaming platforms. The Slovak training sample's cultural and demographic characteristics may contribute to this generalization failure, as personality-music associations can vary significantly across cultural contexts.

5.3 Implications for Music Psychology

These results contribute to ongoing debates about the strength and practical significance of personality-music relationships. While the substantial Openness findings confirm theoretical predictions about aesthetic preferences and cultural engagement, the weak associations for other traits challenge broader claims about music as a "window into personality."

The findings suggest that music preferences may be most informative about specific personality facets rather than comprehensive psychological profiles. Openness encompasses curiosity, creativity, and aesthetic sensitivity—characteristics that logically connect to musical exploration and genre preferences. In contrast, traits like Conscientiousness (organization, discipline, goal orientation) and Agreeableness (interpersonal warmth, cooperation) operate in domains where musical choices may be largely irrelevant.

This specificity has important theoretical implications. Rather than viewing music as a general personality indicator, researchers might conceptualize it as specifically diagnostic of aesthetic and cultural dimensions of individual differences. This more circumscribed view aligns with uses and gratifications theory, which suggests that people use music for different psychological functions—emotional regulation, social connection, identity expression—that may not uniformly reflect personality traits (Chamorro-Premuzic and Furnham, 2007).

The substantial individual variation observed in prediction accuracy also highlights the importance of person-situation interactions in musical behaviour. While some participants showed relatively accurate predictions, others exhibited large systematic errors that suggest music choices are influenced by factors beyond stable personality traits.

5.4 Applications in Creative Industries

The original motivation for this research involved developing personality-aware applications for creative industries, particularly festival scheduling systems. The results suggest that such applications face significant challenges that limit their near-term viability.

From a technical perspective, the prototype successfully demonstrated the feasibility of integrating personality inference with streaming platform data. Users could authenticate through Spotify, receive personality predictions, and view customized recommendations based on their inferred traits. This proof-of-concept illustrates the potential for personality-aware cultural experiences.

However, the poor predictive accuracy raises serious questions about practical deployment. Festival scheduling systems require reliable personality assessments to make meaningful recommendations. With prediction errors averaging 14.2 points on a 120-point scale and systematic biases across traits, such systems risk providing irrelevant or counterproductive recommendations that could diminish rather than enhance user experience.

The systematic overestimation of Openness could lead to over-recommendation of experimental or avant-garde performances to users who prefer mainstream entertainment. Conversely, the severe underestimation of Agreeableness might result in insufficient recommendation of collaborative or community-oriented activities for highly agreeable individuals. These biases could perpetuate stereotypes or limit users' exposure to diverse cultural experiences.

Nevertheless, personality predictions might serve as one component of multi-faceted recommendation systems that incorporate additional signals. Combining personality estimates with explicit user preferences, social network data, past attendance patterns, and contextual information could potentially improve recommendation quality while reducing reliance on potentially inaccurate personality inference.

The ethical implications of personality-based targeting also warrant careful consideration. Even modestly accurate personality predictions could enable sophisticated psychological manipulation or discrimination, requiring robust ethical frameworks before practical deployment.

5.5 Methodological Limitations

Several methodological constraints limit the conclusions that can be drawn from this study, providing important context for interpreting the results and directions for future research.

Training Data Limitations: The Kaggle Young People Survey provided valuable personality-preference data but suffered from several constraints. The Slovak sample limits cultural generalizability, as personality-music associations may vary significantly across cultures, particularly between Western and non-Western contexts. The survey-based personality measures were ad hoc composites rather than validated Big Five instruments, potentially introducing measurement error that attenuates observed associations. Additionally, the self-reported genre preferences may not accurately reflect actual listening behaviour, introducing additional noise into the training process.

Feature Set Constraints: The reliance on genre preferences as the primary predictive features likely underestimated the full potential of music-based personality inference. Spotify's audio features (valence, energy, danceability, acousticness) provide detailed characterizations of musical content that may contain personality signals not captured by broad genre categories. Behavioural metrics such as listening duration, repeat behaviour, discovery patterns, and temporal listening habits could also provide valuable predictive information.

Sample Size for External Validation: The five-participant validation study, while necessitated by Spotify developer constraints, provided insufficient statistical power for robust evaluation. This limitation prevents strong conclusions about real-world model performance and generalizability. Future research would benefit from larger validation samples obtained through production deployment or academic partnerships with streaming platforms.

Cross-Domain Generalization: The training on survey data and testing on behavioural data represents a significant domain shift that may have contributed to poor external validation performance. The gap between self-reported preferences and actual listening behaviour, combined with cultural differences between the Slovak training sample and the validation participants, likely impaired model generalization.

5.6 Ethical Considerations

The development and potential deployment of music-based personality inference systems raises several ethical concerns that merit careful consideration.

Privacy and Consent: Users typically do not expect their listening history to reveal psychological traits, yet even modestly accurate prediction models could enable such inference without explicit consent. The disconnect between user expectations and algorithmic capabilities creates potential for privacy violations and psychological profiling without informed consent. Future systems must implement transparent consent mechanisms that clearly communicate the scope and accuracy of personality inference.

Algorithmic Bias and Fairness: The systematic prediction biases observed in this study—particularly the overestimation of Openness and severe underestimation of Agreeableness—demonstrate the potential for algorithmic discrimination. These biases could disadvantage certain user groups if deployed in commercial contexts, leading to unfair treatment in recommendation systems, advertising, or cultural access.

Psychological Labeling and Manipulation: Even inaccurate personality predictions risk becoming self-fulfilling prophecies or constraining user behaviour through algorithmic

recommendations. Research demonstrating that personality-based targeting can influence behaviour raises concerns about potential manipulation for commercial or political purposes, potentially undermining user autonomy.

5.7 Future Directions

The limitations revealed by this study suggest several promising directions for advancing music-based personality inference research.

Methodological Improvements: Future research should prioritize larger, more diverse training datasets that combine validated personality assessments with actual listening behaviour rather than self-reported preferences. Cross-cultural validation is essential to address the cultural specificity limitations observed in this study. Longitudinal studies examining whether personality-music relationships remain stable over time could improve prediction reliability and theoretical understanding.

Feature Engineering: Expanding beyond genre preferences to incorporate audio features, lyrical content analysis, and behavioural metrics could significantly improve predictive performance. Spotify's extensive audio feature database provides rich characterizations of musical content that may contain personality signals not captured by genre categories.

Advanced Modeling Approaches: Deep learning architectures, particularly multimodal models that integrate audio, text, and behavioural data, show promise for improving prediction accuracy. Ensemble methods combining multiple algorithms might achieve more robust performance across traits.

Hybrid Systems and Applied Validation: Rather than relying solely on personality inference, future applications might combine personality estimates with other signals to improve recommendation quality. Moving beyond academic validation to testing in real-world creative industry applications could provide more realistic assessments of system utility and user acceptance.

This dissertation demonstrates that while music listening patterns contain some personality-related signals, current approaches are insufficient for reliable individual-level trait prediction across most personality dimensions. The findings contribute to personality computing literature by providing an honest assessment of current capabilities and limitations, challenging optimistic claims while confirming specific theoretical relationships. The study illustrates both the promise and the pitfalls of applying computational approaches to psychological assessment, emphasizing the need

for methodological rigor, ethical consideration, and realistic expectations about predictive performance.

6. Conclusion

This dissertation investigated whether personality traits can be inferred from music listening patterns with sufficient accuracy to support creative industry applications, specifically personalised festival scheduling. Using machine learning models trained on the Kaggle Young People Survey dataset and validated through a prototype Spotify application, the study provided both statistical validation and real-world assessment of predictive capabilities.

The results deliver a sobering assessment of current music-based personality prediction capabilities. While some personality signals exist within music listening data, current approaches are insufficient for reliable individual-level trait prediction across most personality dimensions. The findings reveal a stark dichotomy: **Openness to Experience** achieved meaningful predictive validity ($R^2 = 0.36$), explaining 36% of personality variance with statistically significant associations to complex and alternative musical genres. However, the remaining four traits—Conscientiousness, Extraversion, Agreeableness, and Neuroticism—showed negligible explanatory power ($R^2 < 0.1$), often failing to significantly outperform chance predictions.

External validation with five Spotify users exposed even greater limitations, revealing high prediction errors (MAE = 14.2 points on 0-120 scale) and systematic biases that challenge practical deployment. Openness was consistently overestimated, potentially reflecting the model's tendency to misinterpret diverse listening patterns as indicators of high openness. Agreeableness was severely underestimated across nearly all participants, highlighting the theoretical disconnect between interpersonal traits and musical preferences.

Methodological Contributions

This study makes several important contributions to personality computing literature. First, it demonstrates the value of combining statistical validation with real-world testing through prototype deployment. While the Kaggle dataset provided controlled conditions for model training, Spotify validation revealed generalization challenges that pure statistical evaluation might miss.

Second, the research introduced novel **temporal consistency analysis** incorporating Shannon diversity metrics and multi-timeframe listening pattern evaluation. This approach examined how behavioural regularity and musical exploration patterns could theoretically inform personality predictions through genre stability measurements, preference correlations, and listening diversity calculations. However, these sophisticated consistency-based adjustments failed to substantially improve overall prediction accuracy, suggesting that even advanced behavioural analysis cannot overcome fundamental limitations in music-personality associations.

Third, the study provides transparent, reproducible methodology using publicly available datasets and open-source tools, contrasting with research relying on proprietary datasets. The complete research pipeline—from data preprocessing through temporal analysis to prototype deployment—is documented and available for future research.

Fourth, the honest assessment of poor predictive performance contributes to realistic expectations about music-based personality inference capabilities. Rather than focusing solely on statistically significant associations, the study emphasizes effect sizes, prediction errors, and practical utility—metrics essential for real-world deployment but often underemphasized in academic research.

Theoretical Implications

The superior predictability of Openness compared to other traits supports theoretical frameworks emphasizing aesthetic preferences and cultural engagement in personality expression. Individuals high in Openness seek novel experiences, appreciate complexity, and value cultural sophistication—characteristics that naturally manifest in musical choices through preferences for classical, jazz, and alternative genres over mainstream pop and country.

Conversely, the poor predictability of traits like Agreeableness and Conscientiousness suggests these dimensions operate primarily in social and behavioural domains where musical preferences may be largely irrelevant. This challenges broad claims about music as a comprehensive "window into personality," instead supporting a circumscribed view where music primarily reflects aesthetic and cultural dimensions of individual differences.

The substantial individual variation in prediction accuracy highlights person-situation interactions in musical behaviour. While some individuals may express personality consistently through music choices, others' listening patterns are dominated by contextual factors, social influences, or specific uses that obscure stable trait signals.

Implications for Creative Industries

The original motivation involved developing personality-aware applications for creative industries. Results suggest such applications face significant challenges limiting near-term viability. Festival scheduling systems require reliable personality assessments to make meaningful recommendations, but the high prediction errors and systematic biases could lead to irrelevant or counterproductive suggestions.

Nevertheless, the successful technical implementation demonstrates the feasibility of integrating personality inference with streaming platform data. Such systems could potentially serve as one component of multi-faceted recommendation engines combining personality estimates with explicit user preferences, social network data, and contextual information. This hybrid approach could improve recommendation quality while reducing reliance on potentially inaccurate personality inference.

The ethical implications of personality-based targeting require careful consideration. Even modestly accurate personality predictions could enable sophisticated psychological manipulation or discrimination, necessitating robust ethical frameworks before practical deployment.

Limitations and Future Research

Several limitations constrain conclusions. The Slovak training dataset limits cultural generalizability, while five-participant external validation provides insufficient statistical power for robust real-world performance assessment. Reliance on genre preferences rather than detailed audio features or behavioural metrics may have underestimated music-based personality inference potential.

Future research should prioritize larger, more diverse datasets combining validated personality assessments with actual listening behaviour. Cross-cultural validation is essential to address observed cultural specificity limitations. Advanced modeling approaches incorporating multimodal data—audio features, lyrical content, behavioural patterns—could potentially achieve better predictive performance.

The temporal consistency analysis, while methodologically sound, highlighted that behavioural sophistication alone cannot compensate for weak underlying personality-music relationships, suggesting fundamental issues lie in signal strength rather than analytical complexity.

Final Assessment

This dissertation demonstrates that while music listening patterns contain some personality-related signals—particularly for Openness to Experience—current approaches are insufficient for reliable individual-level trait prediction across most personality dimensions. The research contributes to personality computing literature by providing realistic assessment of capabilities and limitations while highlighting the gap between academic findings and practical utility.

The study illustrates both the promise and pitfalls of applying computational approaches to psychological assessment. While successful technical implementation suggests future potential, poor predictive accuracy and systematic biases emphasize the need for methodological rigor, ethical consideration, and realistic expectations about current technological capabilities.

For creative industries, findings suggest personality-aware applications remain challenging to implement effectively with current methods. However, the research provides a foundation for future development by identifying key limitations and suggesting improvement directions. The honest assessment of both capabilities and constraints serves the field better than overly optimistic claims that could lead to premature or inappropriate deployment.

Ultimately, this dissertation contributes to nuanced understanding of personality-music relationships while demonstrating the importance of rigorous evaluation in applied machine learning research. The findings emphasize that the path from statistical significance to practical utility requires careful consideration of effect sizes, prediction errors, systematic biases, and ethical implications—lessons extending well beyond music-based personality inference to the broader field of computational psychology.

Acknowledgment of AI Assistance

This research utilized Large Language Models (LLMs) including ChatGPT and Claude 4 to assist with: (1) code development and debugging for the Spotify application and machine learning pipeline, (2) literature research and citation formatting, (3) dissertation structure and formatting, (4) creation of tables and figures, and (5) data science methodology guidance. All substantive analysis, interpretation, and conclusions remain the author's original work.

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Appendices

Appendix A: Code Repository

Complete source code for data preprocessing, model training, and Spotify application available at: [https://github.com/MisoMisanko/Spotify_scheduler]

Key Files:

- `train_production_models.py`: Model training script using Kaggle dataset
- `production_predictor.py`: Prediction interface for deployed models
- `app.py`: Streamlit web application with Spotify integration
- `models/production_personality_models.pkl`: Trained Ridge regression models
- `data/kaggle_young_people/responses.csv`: Training dataset (publicly available)

Appendix B: Sample Consent Form

RESEARCH PARTICIPATION CONSENT FORM

Study Title: Predicting Personality from Music Listening Patterns

Researcher: Michal Mišanko, MSc Data Science & AI, University of the Arts London

What this study involves: You are invited to participate in research examining relationships between music listening patterns and personality traits. Participation involves:

1. **Spotify Account Connection:** Granting temporary access to your Spotify listening history through official Spotify authentication (OAuth)
2. **Personality Questionnaire:** Completing a validated Big Five personality assessment (10-15 minutes)
3. **Data Usage:** Your listening data will be analyzed to generate personality predictions for research comparison

Data Collection and Privacy:

- **Spotify Data:** We will access your top artists, recent listening history, and saved tracks. We do NOT access private playlists, personal information, or have ongoing access beyond this study
- **Questionnaire Data:** Anonymous personality scores used only for research validation
- **Data Retention:** All personal data deleted within 30 days of study completion. Only anonymized, aggregated results retained
- **Data Security:** All data stored securely and encrypted. No data shared with third parties

Your Rights:

- **Voluntary Participation:** You may withdraw at any time without penalty
- **Data Deletion:** You may request deletion of your data at any point
- **Anonymity:** Your identity will not be linked to any published results
- **Access:** You may request a copy of your data or personality predictions

Potential Risks and Benefits:

- **Minimal Risk:** No psychological or personal risks anticipated
- **Benefits:** Contributing to research on music psychology and receiving personality insights

Contact Information: For questions or concerns, contact:
m.misanko0120191@arts.ac.uk

Consent Statement: I have read and understood the information above. I voluntarily agree to participate in this research study.

- ☐ I consent to Spotify data access for this research
- ☐ I consent to personality questionnaire completion
- ☐ I understand I can withdraw at any time

Participant Signature: _____ **Date:** _____

Researcher Signature: _____ **Date:** _____